

# JEE MAINS GRAND TEST SERIES



**NARAYANA**  
IIT ACADEMY  
INDIA

**40+**  
YEARS  
OF EXCELLENCE

Sec: SR.IIT\_\*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date: 18-01-24

Max. Marks: 300

Name of the Student: \_\_\_\_\_

H.T. NO:

18-01-24\_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)\_JEE-MAIN\_GTM-20(N)\_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

| SUBJECT   | MISTAKES            |                           |              |
|-----------|---------------------|---------------------------|--------------|
|           | JEE<br>SYLLABUS Q'S | JEE<br>EXTRA SYLLABUS Q'S | TOTAL<br>Q'S |
| MATHS     |                     |                           |              |
| PHYISCS   |                     |                           |              |
| CHEMISTRY |                     |                           |              |

**For More Material Join: @JEEAdvanced\_2024**

## PHYSICS

MAX.MARKS: 100

SECTION – I  
(SINGLE CORRECT ANSWER TYPE)

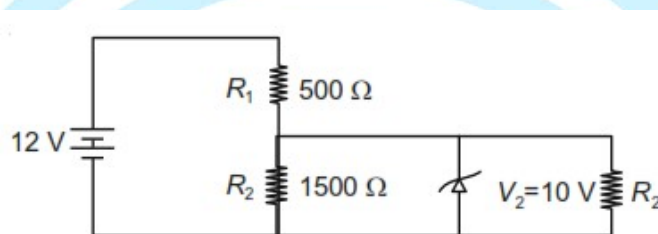
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

1. In  $u - v$  method to find focal length of a concave mirror, the object distance is found to be 10.0 cm and image distance is also found to be 10.0 cm, then find the focal length is

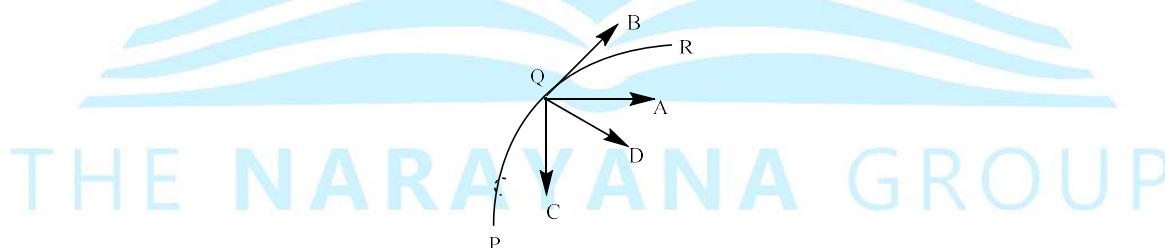
A)  $(5.00 \pm 0.05) \text{ cm}$                       B)  $(5.00 \pm 0.02) \text{ cm}$   
C)  $(5.00 \pm 0.01) \text{ cm}$                       D)  $(5.00 \pm 0.09) \text{ cm}$

2. In the given circuit the current through Zener Diode is close to



A) 6.7 mA                      B) 0.0 mA                      C) 4.0 mA                      D) 6.0 mA

3. A body moves along a path PQR in horizontal plane from P to R. When the particle is at Q, its speed is decreasing. The acceleration of the particle at Q is best represented by the vector. (A, B, C and D are vectors)



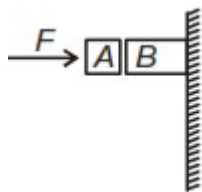
A) A                      B) B                      C) C                      D) D

4. A man holds a ball of mass  $(1/2) \text{ kg}$  in his hand. He throws it vertically upward. During this process his hand moves up by 40 cm and the ball leaves his hand with an upward velocity of  $4 \text{ ms}^{-1}$ . The constant force with which the man pushes the ball is

( $g = 10 \text{ m/s}^2$ )

A) 2N                      B) 10 N                      C) 15 N                      D) 7 N

5.



Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is

- A) 100 N      B) 80 N      C) 120 N      D) 150 N

6. This question has Statement-I and Statement-II. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-I: A point particle of mass  $m$  moving with speed  $v$  collides with stationary point particle of mass  $M$ . If the maximum energy loss possible is given as  $f\left(\frac{1}{2}mv^2\right)$

then  $f = \left(\frac{m}{M+m}\right)$ .

Statement-II: Maximum energy loss occurs when the particles get stuck together as a result of the collision.

- A) Statement-I is true, Statement-II is true, Statement-II is a correct explanation of Statement-I  
 B) Statement-I is true, Statement-II is true, Statement-II is not a correct explanation of Statement-I  
 C) Statement-I is true, Statement-II is false.  
 D) Statement-I is false, Statement-II is true.

7. A thin long smooth tunnel is dug along the diameter of the earth from North pole to South pole. A body starts from rest from the North Pole and travels along the tunnel to the center of the earth. The speed of the body when it reaches the center of the earth is [ $M_e$  is mass of earth and  $R_e$  is radius of earth,  $G$  Universal gravitational constant]

- A) zero      B)  $\sqrt{\frac{3GM_e}{2R_e}}$       C)  $\sqrt{\frac{2GM_e}{3R_e}}$       D)  $\sqrt{\frac{GM_e}{R_e}}$

8. Match column -I with column -II

Column -I

Column -II

p) Equation of continuity

1) Less than critical velocity

q) Bernoulli's theorem

2) Formation of eddies & vortices

r) Turbulent flow

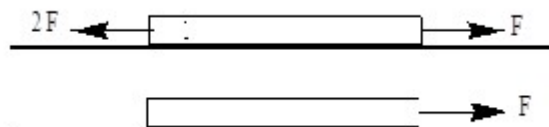
3) Law of conservation of mass

s) Stream line flow

4) Law of conservation of energy

- |    | p | q | r | s |
|----|---|---|---|---|
| A) | 3 | 4 | 2 | 1 |
| B) | 3 | 4 | 1 | 2 |
| C) | 4 | 3 | 2 | 1 |
| D) | 2 | 3 | 1 | 4 |

9. Statement-1: Two identical uniform elastic rods are lying horizontally on smooth horizontal surface under the action of forces as shown in figure. The elongation in two rods will be same.



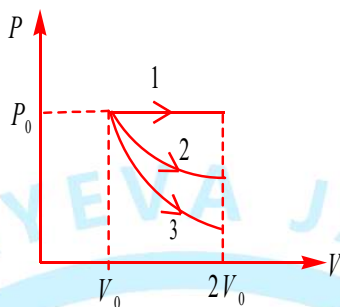
Statement-2: The magnitude of acceleration of rod will be same in both cases.

- A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- C) Statement-1 is true, statement-2 is false.
- D) Statement-1 is false, statement-2 is true.



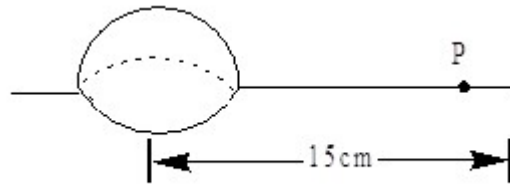
10. A gas is expanded from volume  $V_0$  to  $2V_0$  under three different processes.

Process 1 is isobaric process, process 2 is isothermal and process 3 is adiabatic. Let  $\Delta U_1, \Delta U_2$  and  $\Delta U_3$  be the change in internal energy of the gas in these three processes. Then

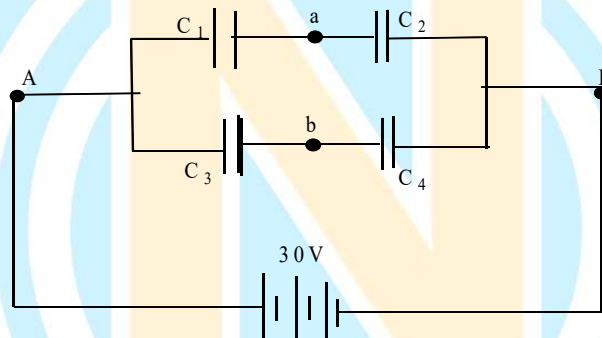


- A)  $\Delta U_1 > \Delta U_2 > \Delta U_3$                       B)  $\Delta U_1 < \Delta U_2 < \Delta U_3$   
 C)  $\Delta U_2 < \Delta U_1 < \Delta U_3$                       D)  $\Delta U_2 < \Delta U_3 < \Delta U_1$
11. The temperature of an ideal gas is increased from 120K to 480K. If at 120K, the root mean square velocity of the gas molecules is  $v$ , then at 480K it becomes  
 A)  $4v$                       B)  $2v$                       C)  $v/2$                       D)  $v/4$
12. A U tube of uniform bore of cross-sectional area  $A$  is set up vertically with open ends up. A liquid of mass  $M$  and density  $d$  is poured into it. The liquid column will oscillate with a period(  $g$  is acceleration due to gravity )  
 A)  $2\pi\sqrt{\frac{M}{g}}$                       B)  $2\pi\sqrt{\frac{MA}{dg}}$                       C)  $2\pi\sqrt{\frac{M}{Adg}}$                       D)  $2\pi\sqrt{\frac{M}{2Adg}}$
13. A tuning fork of frequency 340 Hz is sounded above a cylindrical tube of 1 m high. Water is slowly poured into the tube. If the speed of sound is  $340\text{ms}^{-1}$ , at what levels of water in the tube will the sound of the fork be appreciably intensified?  
 A) 25 cm, 75 cm                      B) 20 cm, 80 cm  
 C) 15 cm, 85 cm                      D) 17 cm, 83 cm

14. Figure shows a solid hemisphere with a charge of  $5\text{nC}$  distributed uniformly throughout its volume. The hemisphere lies on a plane and point P is located on the plane, along a radial line from the centre of curvature at distance  $15\text{ cm}$  from the center. The electric potential at point P due to the hemisphere, is:  $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ (S.I. units)}\right)$

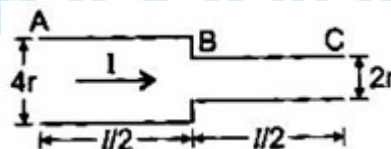


- A) 150 V      B) 300 V      C) 450 V      D) 600 V
15. Four capacitors with capacitances  $C_1 = 1\mu\text{F}$ ,  $C_2 = 1.5\mu\text{F}$ ,  $C_3 = 2.5\mu\text{F}$  and  $C_4 = 0.5\mu\text{F}$  are connected as shown and are connected to a  $30\text{ V}$  source. The potential difference between points a and b is



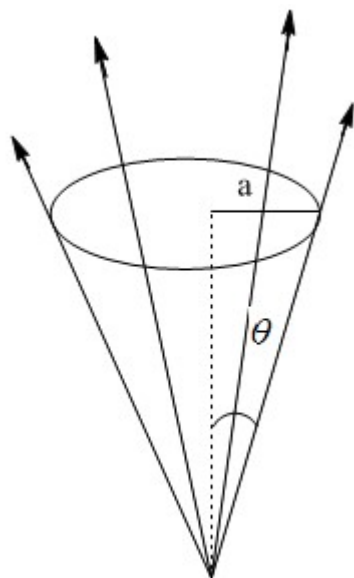
- A) 5V      B) 9V      C) 10 V      D) 13 V
16. Consider a cylindrical element as shown in the adjoining figure. Current flowing through the element is  $I$  and resistivity of the material of the cylinder is  $\rho$ .

Choose the correct option out of the following.



- A) Drift velocity in the second half is equal to that of in the first half
- B) Current density in both halves are equal
- C) Electric field in both halves is equal.
- D) Voltage drop in first half is  $1/4$  of voltage drop in second half.

17. A circular loop carrying current ' $i$ ' of radius  $a$  is placed in a radial magnetic field as shown. The strength of magnetic field at the site of wire is  $B$ . The field lines form a cone of semi vertical angle  $\theta$ . The net force acting on the loop is:



- A) zero      B)  $2\pi Bai \cos \theta$       C)  $2\pi Bai \sin \theta$       D) None of these
18. A rod of ferromagnetic material with dimensions 10 cm x 0.5 cm x 2 cm is placed in a magnetising field of intensity  $2 \times 10^5$  A/m. The magnetic moment produced due to it is  $6 \text{ Am}^2$ . The value of magnetic induction in the rod will be ..... $10^{-2} \text{ T}$ .
- A) 100.5      B) 200.5      C) 50.24      D) 300.5
19. **Directions:** This question consists of two statements, each printed as Assertion and Reason. While answering the question, you are required to choose any one of the following four response.
- Assertion :** Induced emf will always occur whenever there is change in magnetic flux.
- Reason :** Current always induces whenever there is change in magnetic flux.
- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

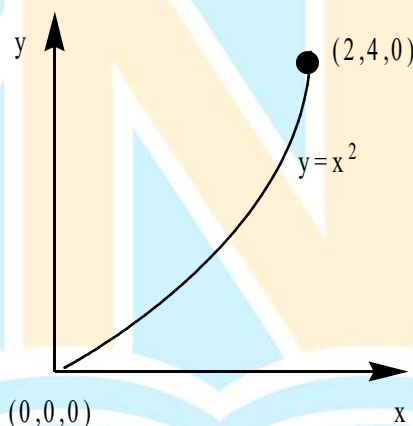
20. Two identical glass  $\left(\mu_g = \frac{3}{2}\right)$  equi-convex lenses of focal length  $f$  are kept in contact. The space between the two lenses is filled with water  $\left(\mu_w = \frac{4}{3}\right)$ . The focal length of the combination
- A)  $f$                       B)  $\frac{f}{2}$                       C)  $\frac{4f}{3}$                       D)  $\frac{3f}{4}$

**SECTION-II**  
**(NUMERICAL VALUE ANSWER TYPE)**

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

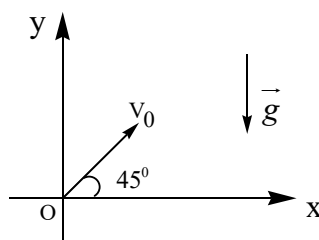
**Marking scheme: +4 for correct answer, -1 in all other cases.**

21. By applying a force  $\vec{F} = (3xy - 5z)\hat{j} - 4z\hat{k}$  a particle is moved along the path  $y = x^2$  from point  $(0,0,0)$  to the point  $(2,4,0)$ . The work done by the  $F$  on the particle in joule is  $X/5$ . The value of  $X$  is



22. A billiard ball has mass 250 gm and radius 2.5cm and is initially at rest. A horizontal rod is used at a height  $h$  vertically above centre  $C$  to hit the ball. The ball immediately begin to roll without slipping. Find the value of  $h$  in mm
23. A particle is projected at time  $t = 0$  from origin  $O$  with a speed  $V_0$  at an angle  $45^\circ$  to the horizontal. The angular momentum about origin after time  $t = \frac{V_0}{g}$  is  $\frac{3mV_0^3}{x\sqrt{2}g}$ .

Find the value of  $x$ . ( $g$  is acceleration due to gravity)





24. Two metal spheres are falling through a liquid of density  $2 \times 10^3 \text{ kg/m}^3$  with the same terminal speed. The material density of sphere 1 and sphere 2 are  $8 \times 10^3 \text{ kg/m}^3$  and  $11 \times 10^3 \text{ kg/m}^3$  respectively. The ratio of their radii is  $\sqrt{Q}$ . The value of  $10Q$  is
25. Two sitar strings "A" and "B" playing the note "Dha" are slightly out of tune and produce beats of frequency 5 Hz. The tension of the string "B" slightly increased and the beat frequency is found to decrease to 3 Hz. What is the original frequency of "B" in hertz, if the frequency of "A" is 427 Hz?
26. A capacitor of capacitance  $1 \mu\text{F}$  can withstand the maximum voltage 6 kV while a capacitor of  $2 \mu\text{F}$  can withstand the maximum voltage 4 kV. What maximum voltage (in kV) will the system of these two capacitors withstand, if they are connected in series?
27. An electric bulb rated 100 W- 300 V DC is connected with an AC supply of 500 V and  $\frac{150}{\pi}$  Hz. The required inductance in henry, to be connected in series to the bulb so that the bulb works at the rated power is.....
28. An electromagnetic wave of intensity  $1400 \text{ W/m}^2$  falls on a metal surface of area  $1.5 \text{ m}^2$  is completely absorbed by it. The force exerted by the beam is  $p \times 10^{-6} \text{ N}$ . The value of  $p$  is
29. In a Young's double slit experiment, the slit widths are different such that the ratio of slit widths is 4. The ratio of maximum intensity and minimum intensity in the fringe pattern obtained is \_\_\_\_\_
30. An electromagnetic radiation whose electric component varies with time as  $E = C_1(C_2 + C_3 \cos \omega t) \cos \omega_0 t$ , here  $C_1, C_2$  and  $C_3$  are constants, is incident on lithium and liberates photoelectrons. If the kinetic energy of the most energetic electron is 2.6 eV, the work function of lithium is (in eV), [Take :  $\omega_0 = 2.4\pi \times 10^{15} \text{ rad/s}$  and  $\omega = 8\pi \times 10^{14} \text{ rad/sec}$ , Planck's constant is  $h = 6.6 \times 10^{-34} \text{ MKS}$ ]

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

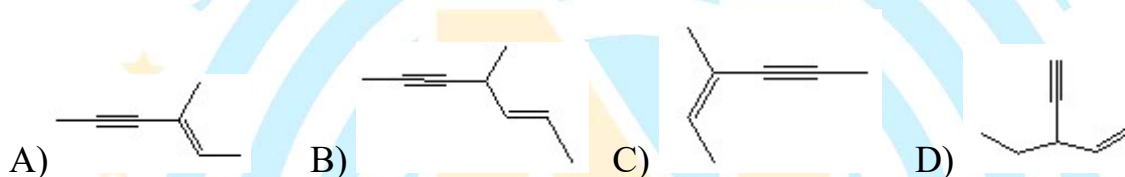
31. The atomic masses of Helium and Neon are 4.0 and 20.0 amu respectively. The value of the de-Broglie wavelength of helium gas is M times the de-Broglie wavelength of neon. The value of M is (The ratio of kinetic energy of He and Ne given as 1 : 5)

A) 5                      B) 25                      C) 1/5                      D) 1/25

32. The magnetic moment of an octahedral Co(II) complex is 4.0 B.M (nearly) The electronic configuration of the complex is

A)  $t_{2g}^5 e_g^2$                       B)  $t_{2g}^6 e_g^1$                       C)  $t_{2g}^3 e_g^4$                       D)  $t_{2g}^4 e_g^3$

33. The ene-yne that produces a chiral compound upon treatment with Lindlar Catalyst is



34. The dipole moment of HCl is 1.08D and the bond distance is 1.27 Å, then the fraction of charge on hydrogen and chlorine, respectively, relative to charge of electron

A) +1.0 and -1.0                      B) +0.85 and -0.85  
C) +0.356 and -0.356                      D) +0.178 and -0.178

35. The correct order of ionic radii of Ce, La, Pm, and Yb in +3 oxidation state is

A)  $Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$                       B)  $La^{3+} < Pm^{3+} < Ce^{3+} < Yb^{3+}$   
C)  $La^{3+} < Ce^{3+} < Pm^{3+} < Yb^{3+}$                       D)  $Yb^{3+} < Ce^{3+} < Pm^{3+} < La^{3+}$

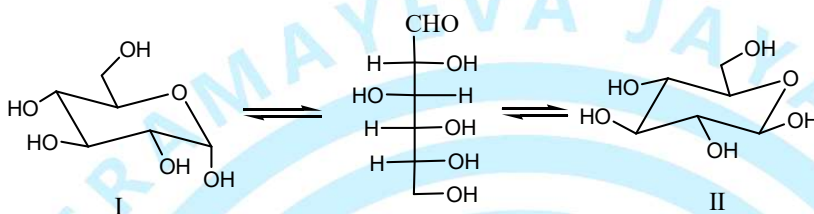
36. 27. At 35°C, the vapour pressure of  $CS_2$  is 512 mm Hg and that of acetone is 344 mm Hg. A solution of  $CS_2$  in acetone has a total vapour pressure of 600mm Hg. The false statement amongst the following is

A) Raoult's law is not obeyed by this system  
B) A mixture of 100 mL  $CS_2$  and 100 mL acetone has a volume < 200 mL  
C) Heat must be absorbed in order to produce the solution at 35°C  
D)  $CS_2$  and acetone are less attracted to each other than to themselves  
A) Only III                      B) Both I and III                      C) Both I and II                      D) Both II and III

37. For adiabatic reversible expansion of a monoatomic gas (M) and a diatomic gas (D) at initial temperature  $T_i$ , has been carried out independently from initial volume  $V_1$  to final volume  $V_2$ . The final temperature ( $T_M$  for monoatomic and  $T_D$  for diatomic) attained will be

- A)  $T_M = T_D > T_i$       B)  $T_M < T_D < T_i$       C)  $T_M > T_D > T_i$       D)  $T_D < T_M < T_i$

38. Specific rotation of freshly prepared aqueous solution of **I** and **II** are +112 and +18.7, respectively. On standing the optical rotation of aqueous solution of **I** slowly decreases to give a final value of +52.7 due to equilibration with **II**. Under this state of equilibrium, what is the ratio of **II**:**I**?

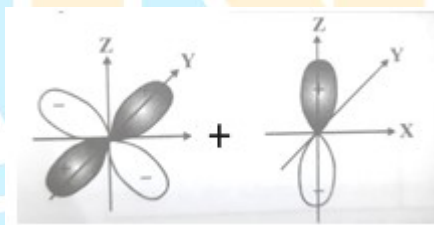


- A) 0.57      B) 1.00      C) 1.75      D) 5.9

39. The half life of a first order reaction varies with temperature according to

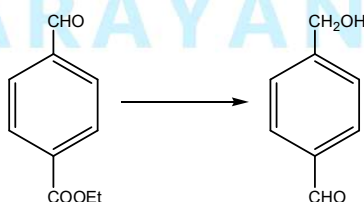
- A)  $\ln t_{1/2} \propto 1/T$       B)  $\ln t_{1/2} \propto T$       C)  $t_{1/2} \propto 1/T^2$       D)  $t_{1/2} \propto T^2$

40. The overlap between the atomic orbitals sketched below is



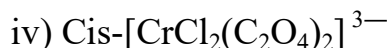
- A) Positive      B) negative      C) zero      D) No overlap

41. Identify the correct reagents required for the following transformation



- A) (i)  $\text{NaBH}_4$ ; (ii)  $\text{H}_3\text{O}^+$   
 B) (i)  $\text{LiAlH}_4$ ; (ii)  $\text{H}_3\text{O}^+$   
 C) (i)  $\text{HOCH}_2\text{CH}_2\text{OH}$ ,  $\text{H}^+$ ; (ii)  $\text{LiAlH}_4$ ; (iii)  $\text{H}_3\text{O}^+$   
 D) (i)  $\text{HSCH}_2\text{CH}_2\text{SH}$ ,  $\text{H}^+$ ; (ii)  $\text{Ni}/\text{H}_2$ ; (iii)  $\text{H}_3\text{O}^+$

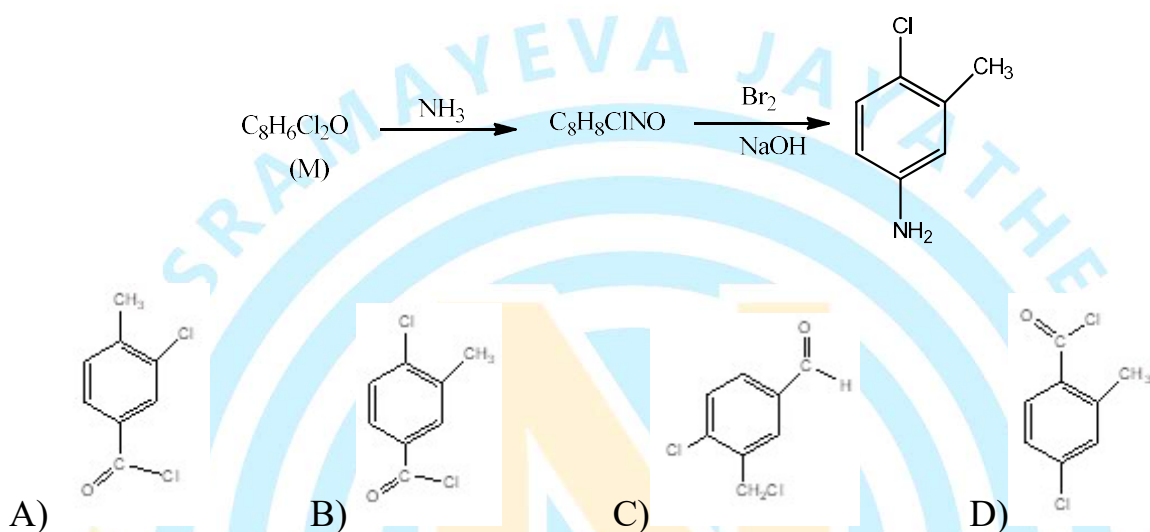
42. Among the following complexes,



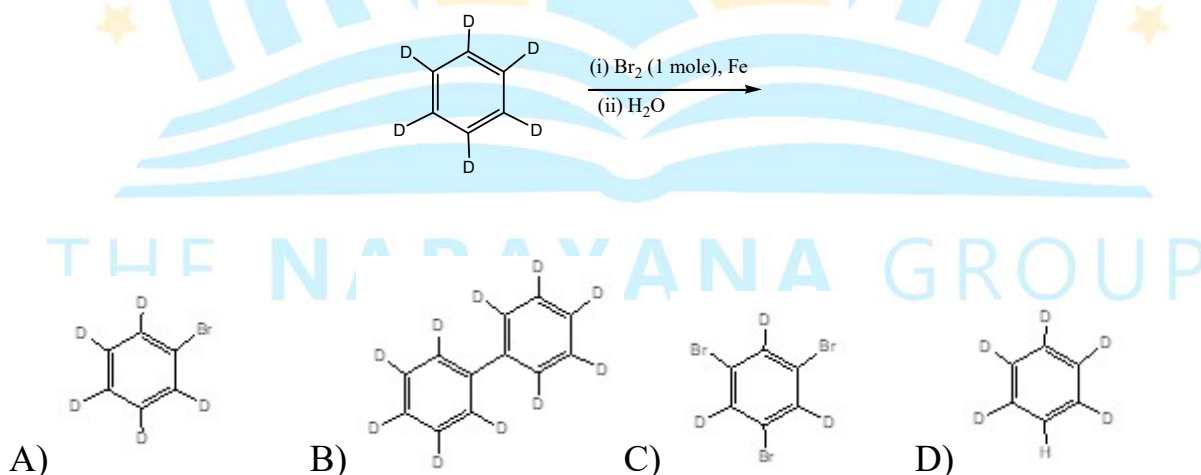
The one that show chirality are

- A) (i), (ii), (iv)      B) (i), (ii), (iii)      C) (ii), (iii), (iv)      D) (i), (iii), (iv)

43. Identify 'M' in the following sequence of reactions



44. The major product(s) obtained from the following reaction of 1 mole of hexadeuterio benzene is/are



45. A given amount of  $\text{Fe}^{2+}$  is oxidized by 'x' mol of  $\text{MnO}_4^-$  in acidic medium. The number of moles of  $\text{Cr}_2\text{O}_7^{2-}$  required to oxidize the same amount of  $\text{Fe}^{2+}$  in acidic medium is

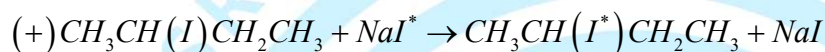
- A) x      B) 0.83 x      C) 2.0 x      D) 1.2 x



46. 250 mL of a waste solution obtained from the workshop of a goldsmith contains 0.1 M  $AgNO_3$  and 0.1 M  $AuCl$ . The solution was electrolyzed at 2V by passing a current of 1A for 15 minutes. The metal/metals electrodeposited will be

$$\left[ E_{Ag^+/Ag}^0 = 0.80V, E_{Au^+/Au}^0 = 1.69V \right]$$

- A) Silver and gold in equal mass proportion  
B) Silver and gold in proportion to their atomic weights  
C) Only gold  
D) Only silver
47. The reaction of (+) 2-iodobutane and  $NaI^*$  ( $I^*$  is radioactive isotope of iodine) in acetone was studied by measuring the rate of racemization ( $k_r$ ) and the rate of inversion ( $k_i$ ).



For this reaction, the relationship between  $k_r$  and  $k_i$

- A)  $k_i = 2k_r$       B)  $k_i = 0.5k_r$       C)  $k_i = k_r$       D)  $k_i = 0.33k_r$
48. In the Lassaigne's test for the detection of nitrogen in an organic compound, the appearance of blue coloured compound is due to
- A) ferric ferricyanide      B) ferrous ferricyanide  
C) ferric ferrocyanide      D) ferrous ferrocyanide
49. In the following reaction, The product P is



- A)  $RCHO$       B)  $R_2CHOEt$       C)  $R_3CH$       D)  $RCH(OEt)_2$
50. Assertion (A) :  $PCl_5$  exists but  $NCl_5$  does not exist.  
Reason (R) : d – orbital in P – atom takes part in chemical bonding; which is lacking in N – atom
- A) Both A and R are true and R is the correct explanation of A  
B) Both A and R are true but R is not the correct explanation of A  
C) A is true but R is false  
D) A is false but R is true

**SECTION-II**  
**(NUMERICAL VALUE ANSWER TYPE)**

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

**Marking scheme: +4 for correct answer, -1 in all other cases.**

51.  $K_{sp}$  of solution  $BaSO_4$  at  $25^\circ C$  is  $4 \times 10^{-5}$ . The solubility of  $BaSO_4$  in 0.1 M  $Na_2SO_4$  at this temperature will be  $P \times 10^{-4}$  M. then the P is equal to
52. The charge carried by 1 millimole of  $M^{n+}$  ions is 193 coulombs. The value of n is
53. In the study of the water gas reaction  $CO_{2(g)} + H_{2(g)} \rightleftharpoons CO_{(g)} + H_2O_{(g)}$ , a mixture of  $CO_2$  and  $H_2$  initially containing 40 mol%  $H_2$  was brought to equilibrium in a closed vessel. At equilibrium, there was 15% of  $H_2$ . The value of  $K_p$  is .....
54. The enthalpies of combustion of C(graphite) and C(diamond) are  $-393.5 \text{ kJ mol}^{-1}$  and  $-395.3 \text{ kJ mol}^{-1}$ , respectively. The enthalpy change of the reaction  $C(\text{graphite}) \rightarrow C(\text{diamond})$  is  $P \times 10^2 \text{ J mol}^{-1}$ . Then P is...
55. The dihedral (torsional) angle (in degrees) between the two methyl groups in the most stable conformation of *n*-butane is .....(molecule is observed through C-2 and C-3 bond)
56. The total number of all possible isomers of  $[Co(H_2NCH_2CH_2NH_2)_2Cl_2]^+$  and  $[Co(H_2NCH_2CH_2NH_2)_3]^{+3}$  together is....
57. The number of lone pairs present in  $XeOF_4$  is...
58. How many structural isomers are  $C_4H_{10}O$  react with  $CH_3MgBr$  to evolve  $CH_4$  gas ?
59. The number of hydrogen bond(s) present in a guanine-cytosine base pair is....
60. Among  $Zn(OH)_2$ ,  $Sn(OH)_2$ ,  $Cu(OH)_2$ ,  $Fe(OH)_3$ ,  $Pb(OH)_2$ ,  $Al(OH)_3$   
The number of hydroxides which are soluble in excess of NaOH solution is

## MATHEMATICS

MAX.MARKS: 100

## SECTION – I

## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

61. Let  $f(x) = \frac{(x^2 - 1)(x^2 - 9)}{k}$ ;  $x \in \mathbb{R}$ ,  $k, n \in \mathbb{N}$ , and if 'A' denotes the set of ordered pairs  $(n, k)$  where  $|f(x)| = n$  has exactly '4' distinct real roots, 'B' denotes the set of ordered pairs  $(n, k)$  where  $|f(x)| = n$  has exactly '7' distinct real roots. Then cardinality of  $A \times B$  is \_\_\_\_\_
- A) 24                      B) 25                      C) 15                      D) 10
62. Two distinct complex numbers  $z_1$  &  $z_2$  satisfy the equation  $z + 1 + 8i = |z|(1 + i)$ , then  $|z_1| + |z_2| =$  \_\_\_\_\_
- A) 16                      B) 18                      C) 13                      D)  $8\sqrt{2}$
63. Let A be the set of natural numbers which are multiples of 5 strictly less than 100, and B be the set of natural numbers which divide 100. The cardinalities of the following sets  $B \setminus A$  (the set of elements in B but not in A),  $A \cap B$ , and B are p, m, n respectively. Then  $p + m + n =$  \_\_\_\_\_
- A) 20                      B) 16                      C) 18                      D) 14
64. The number of real value of 'x' for which
- $$\begin{vmatrix} x^3 - x - 1 & 2x^3 & 2x^3 \\ 2x & x - 1 - x^3 & 2x \\ 2 & 2 & 1 - x - x^3 \end{vmatrix} = 0$$
- is \_\_\_\_\_
- A) 1                      B) 2                      C) 3                      D) 4
65. Words with or without meaning) of 8 letters are formed using the letters of the word EQUATION. If the number of such words where exactly one letter is repeated twice and no other letter is repeated, is  $k \cdot 8!$  then  $k =$  \_\_\_\_\_
- A) 38                      B) 28                      C) 20                      D) 22

66. For  $n = 1, 2, 3, \dots, 50$ . Let  $A = \left\{ a_n / a_n = \begin{cases} (-1)^{\frac{n}{2}} \left( \frac{n}{2} \right), & \text{if } n \text{ is even} \\ (-1)^{\frac{n-1}{2}} \left( \frac{n-1}{2} \right), & \text{if } n \text{ is odd} \end{cases} \right\}$  and B is

the set of all distinct elements of A. The number of permutations of all the elements of set B such that even integers are in increasing order, is

- A)  $\frac{26!}{12!}$       B)  $\frac{26!}{13!}$       C)  $\frac{50!}{24! 26!}$       D)  $\frac{26!}{13! 12!}$

67.  $\lim_{x \rightarrow 2} \frac{\sqrt{(x+7)} - 3\sqrt{(2x-3)}}{\sqrt[3]{(x+6)} - 2\sqrt[3]{3x-5}} =$

- A)  $\frac{31}{20}$       B)  $\frac{34}{23}$       C)  $\frac{23}{31}$       D)  $\frac{39}{17}$

68. Let  $f(x) = \begin{cases} 4x^2 + 2[x]x & \text{if } -\frac{1}{2} \leq x < 0 \\ ax^2 - bx & \text{if } 0 \leq x < \frac{1}{2} \end{cases}$  where  $[x]$  denotes the greatest integer function.

Then

- A)  $f(x)$  is continuous and differentiable in  $\left(-\frac{1}{2}, \frac{1}{2}\right)$  for all real  $a$ , provided  $b = 2$   
 B)  $f(x)$  is continuous and differentiable in  $\left(-\frac{1}{2}, \frac{1}{2}\right)$  only if  $a = 4, b = 2$   
 C)  $f(x)$  is continuous and differentiable in  $\left(-\frac{1}{2}, \frac{1}{2}\right)$  if  $a = 4$  and  $b = 0$   
 D) There exists no real  $a$  and  $b$  such that  $f(x)$  is differentiable in  $\left(-\frac{1}{2}, \frac{1}{2}\right)$

69. If  $x \in \mathbb{R}^+$  and  $n \in \mathbb{N}$ , we can uniquely write  $x = mn + r$  where  $m \in \mathbb{W}$  and  $0 \leq r < n$ .

We defined  $x \bmod n = r$  for example  $10.3 \bmod 3 = 1.3$ . The number of points of discontinuity of the function  $f(x) = (x \bmod 2)^2 + (x \bmod 4)$  in the interval  $0 < x < 9$  is

- A) 4      B) 2      C) 6      D) 8



70. If  $e^y + xy = e$ , then the ordered pair  $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$  at  $x = 0$  is equal to
- A)  $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$       B)  $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$       C)  $\left(\frac{1}{e}, \frac{1}{e^2}\right)$       D)  $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$
71. A particle is moving on the curve with the equation  $y = 2\sin^{-1}(3x)$ . When  $y = \frac{\pi}{3}$  the rate of change of "y" w.r.t time is 2 unit/sec. What is the rate of change of "x" coordinate of the particle w.r.t time at that instant is \_\_\_\_\_ unit/sec
- A)  $\sqrt{3}$       B)  $\frac{1}{\sqrt{6}}$       C)  $\frac{1}{\sqrt{12}}$       D)  $\frac{3}{\sqrt{6}}$
72. Let  $g(x) = a_0 + a_1x + a_2x^2 + a_3x^3$  and  $f(x) = \sqrt{g(x)}$ ,  $f(x)$  has its non-zero local minimum and maximum values at -3 and 3, respectively. If  $a_3 \in$  the domain of the function  $h(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ . The value of  $a_1 + a_2 =$
- A) 30      B) 26      C) 27      D) -27
73. Area traced by the region "M" where  $M = \left\{(x, y) / y \leq 1 + \sqrt{x}, x \geq 0, y \leq 4^{\frac{x}{9}}\right\}$  is \_\_\_\_\_
- A)  $27\log_4\left(\frac{4}{e}\right)$       B)  $27\ln\left(\frac{4}{e}\right)$       C)  $3\ln\left(\frac{27}{e}\right)$       D)  $15\ln\left(\frac{4}{e}\right)$
74.  $\int \frac{x^9}{(x^2 + 4)^6} dx$  is equal to
- A)  $\frac{1}{5x} \left(4 + \frac{1}{x^2}\right)^{-5} + c$       B)  $\frac{1}{5} \left(4 + \frac{1}{x^2}\right)^{-5} + c$
- C)  $\frac{1}{10x} (1 + 4x^2)^{-5} + c$       D)  $\frac{1}{40} (1 + 4x^{-2})^{-5} + c$

75. A function  $y = f(x)$  satisfies  $x \frac{dy}{dx} = \sqrt{y^2 + 1}$ ,  $x > 0$  &  $f(2) = 0$ , then  $f(4) =$  \_\_\_\_\_
- A) 1                      B) 0.75                      C) 0.25                      D) 0.5
76. Locus of centre of circle touching perpendicular lines from origin and the line  $x + y = 1$  is \_\_\_\_\_
- A)  $2xy - 2x - 2y + 1 = 0$                       B)  $2x^2 + 2y^2 - x - y = 0$
- C)  $2xy + 2x + 2y + 1 = 0$                       D)  $x^2 + y^2 - 2x - 2y + 1 = 0$
77. The sides of  $\Delta ABC$  are  $2x - y + 5 = 0$ ,  $x + y - 5 = 0$  &  $x - 2y - 5 = 0$ . Sum of the tangents of its interior angle is
- A) 6                      B)  $\frac{27}{4}$                       C) 9                      D)  $-\frac{3}{4}$
78. The point  $P(ap^2, 2ap)$  lies on the parabola  $y^2 = 4ax$ , where  $a$  is a positive real constant.  $F$  is focus & 'O' is origin. Straight line which passes through  $P$  &  $F$  meets the directrix of the parabola at  $Q$  so that area of triangle  $OPQ$  is  $\frac{15}{4}a^2$ .
- Then the absolute value of the integral value of 'p' is \_\_\_\_\_
- A) 0                      B) 1                      C) 2                      D) 3
79. The straight lines  $\ell_1$  and  $\ell_2$  have the following vector equations  $\vec{r} = 7\hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + a\hat{k})$  and  $\vec{r} = 3\hat{i} + b\hat{j} + 5\hat{k} + \mu(\hat{i} - \hat{k})$  where  $\lambda, \mu$  are real parameters. Acute angle between  $\ell_1$  &  $\ell_2$  is  $60^\circ$  at their point of intersection. Then maximum value of  $a + b$  is \_\_\_\_\_
- A) 4                      B)  $\frac{32}{5}$                       C)  $\frac{12}{5}$                       D)  $\frac{33}{5}$
80. Equation of the line of shortest distance from  $P(\vec{p})$  to the line  $\vec{r} = \vec{a} + t\vec{b}$  is \_\_\_\_\_ (t, s are parameters)
- A)  $\vec{r} = \vec{p} + s((\vec{p} \times \vec{b}) \times \vec{b} - (\vec{a} \times \vec{b}) \times \vec{b})$                       B)  $\vec{r} = \vec{p} + s(\vec{p} \times \vec{b} + \vec{p} \times \vec{a})$
- C)  $\vec{r} = \vec{p} + s(\vec{p} \times \vec{b} + \vec{b} \times \vec{a})$                       D)  $\vec{r} = \vec{p} + s(\vec{a} \times \vec{b} + \vec{a} + \vec{b})$

**SECTION-II**  
**(NUMERICAL VALUE ANSWER TYPE)**

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

**Marking scheme: +4 for correct answer, -1 in all other cases.**

81. If 'A' is a square matrix of order less than 4 such that

$$|\text{adj}(A) - \text{adj}(A^T)| \neq 0 \text{ and } B = 4\text{adj}(A), \text{ and if } |A| = 4, \text{ then}$$

$$|\text{adj}(AB^2A^{-1}B)| + |AB^2 - B^2A| + |\text{adj}(\text{adj}(B))| = \lambda, \text{ then units digits of } \lambda$$

is \_\_\_\_\_

82. If principle argument of a complex number lies in  $[-\pi, \pi)$ , the principal argument of the

$$\text{complex number } "2\sin\frac{6\pi}{5} + i\left(2 + 2\cos\frac{6\pi}{5}\right)" \text{ is } "\frac{\lambda\pi}{10}" \text{ then } |\lambda| =$$

83. Consider  $I_{n,m} = \int_0^1 \frac{x^n}{x^m - 1} dx$  and  $J_{n,m} = \int_0^1 \frac{x^n}{x^m + 1} dx \quad \forall n > m \text{ and } n, m \in \mathbb{N}.$

$$\text{Let } A = \begin{bmatrix} J_{6,5} & 72 & J_{11,5} \\ J_{7,5} & 63 & J_{12,5} \\ J_{8,5} & 56 & J_{13,5} \end{bmatrix} \text{ and } B = \begin{bmatrix} I_{6,5} & 72 & I_{11,5} \\ I_{7,5} & 63 & I_{12,5} \\ I_{8,5} & 56 & I_{13,5} \end{bmatrix},$$

then find the value of  $\det(A) - \det(B)$

84. If K is a non zero constant real number and  $(2 + kx)^6 = a + bx + bx^2 + cx^3 + \dots$  then

$$[a + b + c] = \underline{\hspace{2cm}} \quad ([.] \text{ is G.I.F})$$

85. Suppose that  $x, x^2 - 24$  and  $x + x^2$  are consecutive terms in arithmetic progression, the sum of all possible values of the common difference of sequence is

86. If  $\alpha = \sum_{n=1}^{\infty} \tan^{-1} \left\{ \frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16} \right\}$  and  $\sin \alpha = \frac{a}{b}$ , where  $a, b \in \mathbb{N}$ , then the least

value of  $(a + b)$  is

87. "Ishan" opens a bag which contains 8 orange flavoured sweets & 12 lemon flavoured sweets. 2 of his friends pick a sweet each at random from the bag and then Ishan takes a random sweet for himself. Probability that Ishan chose a lemon flavoured sweet is  $M$ , then  $10M =$  \_\_\_\_\_
88. The probability of 'Sandesh' being late to school is 0.1. The last 20 days that sandesh went to school are examined. Then the mean and variance of the number of days that sandesh was late in those 20 days are  $x$  &  $y$  respectively then  $10xy =$  \_\_\_\_\_
89.  $\frac{x^2}{2} - \frac{y^2}{4} = 3$ . Locus of midpoints of focal chord for it is a curve 'C'. Eccentricity for 'C' is  $\lambda$ , then  $\lambda^2 =$  \_\_\_\_\_
90. Fundamental period of the function  $f(x) = \frac{\sin 4x - 8\sin^3 x \cos x}{\sin x} - 4\cos 3x + \sin 4x$ . is  $\frac{\lambda\pi}{2}$ , then  $\lambda =$  \_\_\_\_\_

